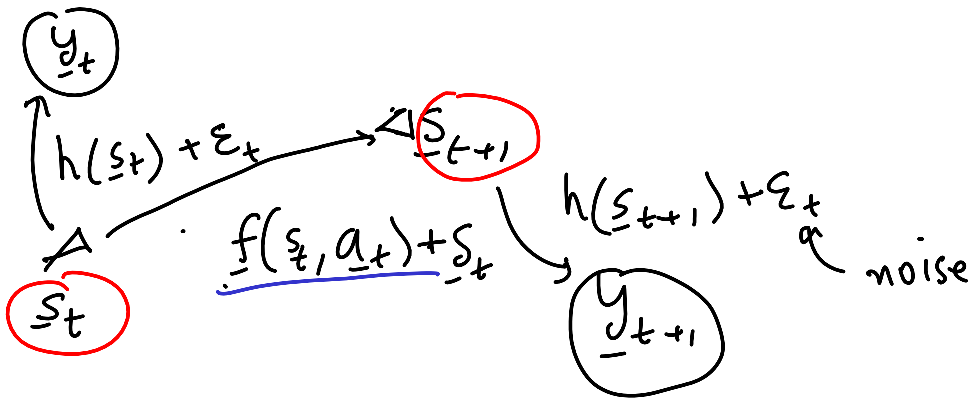
Localization and Mapping

State estimation



We do not know s_{t+1}

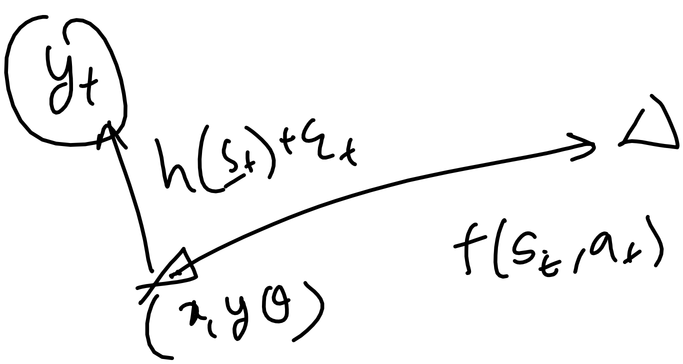
Using the observations, find $\underline{s}_{t+1}$

If we know $\underline{s}_t$ and $f(\underline{s}_t,\underline{a}_t)$, then

$$\underline{s}_{t+1} = f(\underline{s}_t,\underline{a}_t) + \underline{\delta}_t$$

noise

State estimation is the problem of estimating state from observations (noisy) and noisy motion model.



ARUCO markers can give you full state estimate in single observation.

Full state observability

Partial state observability examples:

1. Object detection with camera (given the angle of an object w.r.t the camera but not the distance or the orientation of the object)

2. Laser scan will provide angle and distance of surfaces from the laser scan but not their orientation

$$\underline{y}_t = \underline{h}(\underline{s}_t) + \underline{\varepsilon}_t \quad , \quad \underline{\varepsilon}_t \sim \mathcal{N}(\underline{0}_t, R_t)$$

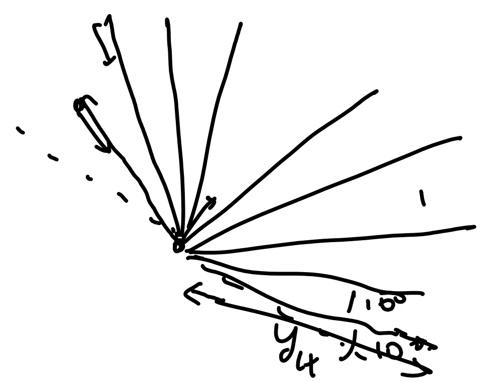
noise

observation vector

observation mode

Gaussian Random vector

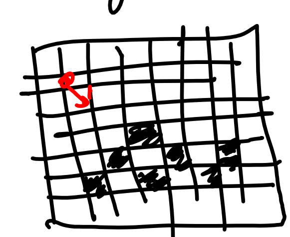
Covariance matrix



$$\underline{y}_t = \begin{bmatrix} y_{1t} \\ \\ \\ \\ y_{nt} \end{bmatrix}$$

$$y_{1t} = \begin{cases} -1 & \text{if it is not reflected} \\ \text{distance from the nearest surface} & \text{if it is reflected} \end{cases}$$

A simple way of representing environment is called occupancy grid: each cell is either occ or free



$$\underline{y}_t = \underline{h}(\underline{s}_t) + \varepsilon_t$$

in full observability $\qquad$ h is invertible

$$\underline{s}_t \approx \underline{h}^{-1}(\underline{y}_t)$$

in partial observability $\qquad$ h is not invertible

Noise is stochastic / probabilistic, we use probabilistic tools

Equiv

$$\underline{s}_{t+1} = \underline{f}(\underline{s}_t, a_t) + \underline{\delta}_t \,, \qquad \underline{\delta}_t \sim \mathcal{N}(\underline{0}, Q_t)$$

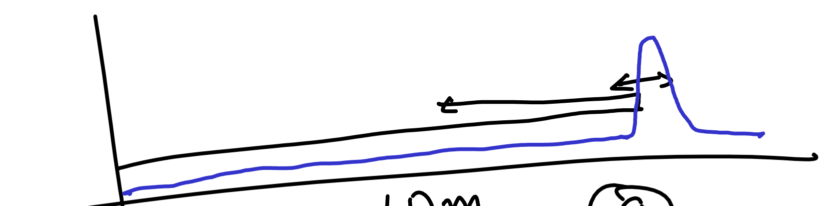
$$P(\underline{s}_{t+1} \mid \underline{s}_t, \underline{a}_t)$$

Transition probability, Motion model

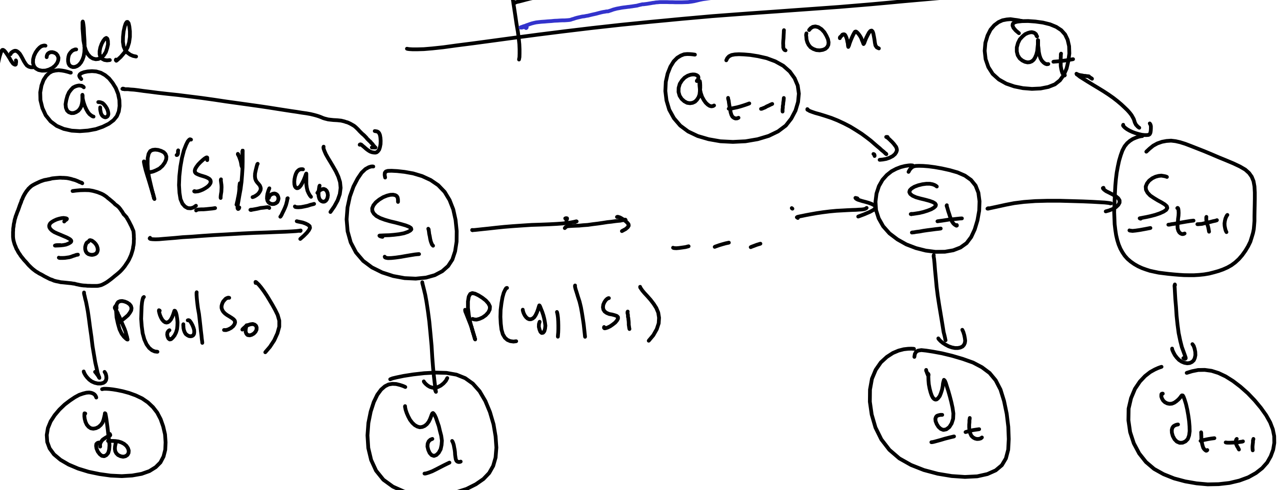
Equiv

$$\underline{y}_t = \underline{h}_t(\underline{s}_t) + \underline{\varepsilon}_t \,, \qquad \varepsilon_t \sim \mathcal{N}(0, R_t)$$

$$P(\underline{y}_t \mid \underline{s}_t)$$



Prob.
Graphical model



each node is a Random Variable, each edge is a Prob. dist.

Markov Decision Process

$$\left(S, A, P(\underline{s}_{t+1} \mid \underline{s}_t, \underline{a}_t), P(\underline{y}_t \mid \underline{s}_t), c(\underline{s}_t, \underline{a}_t)\right)$$

Main assumption: States are Markovian

$$\left(\underline{s}_{t+1} \perp \underline{s}_{t-1}, \underline{s}_{t-2}, \underline{s}_{t-3} \dots \underline{s}_0\right) \mid \underline{s}_t$$

state captures all the historical informations

Formally, we want to estimate

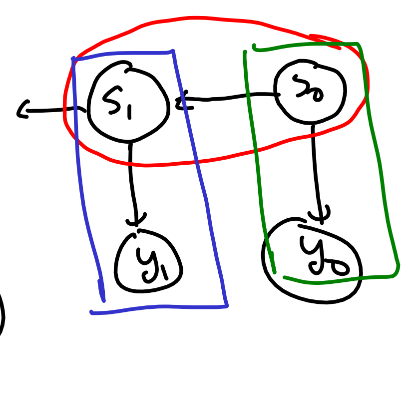
$$P(\underline{s}_t \mid \underline{y}_t, \underline{y}_{t+1}, \dots \underline{y}_0, \underline{a}_t, \underline{a}_{t-1}, \dots, \underline{a}_0)$$

$P(\underline{y}_0 \mid \underline{s}_0)$ = observation model

$$P(\underline{y}_1, \underline{y}_0 \mid s_0) = \int_{s_1} P(\underline{y}_1, \underline{y}_0, \underline{s}_1 \mid s_0) \, ds_1 \longleftarrow \text{Marginalizat}$$

$$= \int_{s_1} \underbrace{P(y_1 \mid \underline{s}_1, \underline{s}_0)}\, P(\underline{s}_1 \mid \underline{s}_0)\, P(\underline{y}_0 \mid \underline{s}_0)\, ds_1$$

$$= \int_{s_1} \boxed{P(\underline{y}_1 \mid \underline{s}_1)}\, \boxed{P(\underline{s}_1 \mid s_0)}\, \boxed{P(\underline{y}_0 \mid s_0)}\, ds_1$$



Probabilistic Theorems

① Marginalization

$$P(x) = \sum_y P(x, y) = \int_y P(x, y)\, dy \quad \Big| \quad P(x \mid z) = \sum_y P(x, y \mid z)$$

② conditional independence

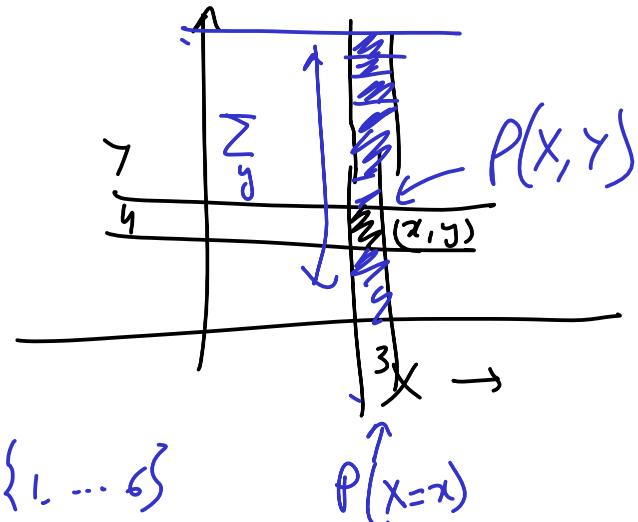
$$P(X, Y \mid Z) = P(X \mid Z)\, P(Y \mid Z) \iff X \perp Y \mid Z$$

$\underbrace{X \text{ and } Y}$ are mutually independent given Z

① Marginalization

Joint probability distribution

$$P(X=x, Y=y)$$



$$P(X=x) = \sum_y P(X=x, Y=y)$$

X is rolling a die 6-faces {1 ... 6}

Y " " " " " {1, ... 6}

② Independence

$$P(X \mid Y) := \frac{P(X,Y)}{P(Y)}$$

definition of conditional prob.

as if Y has occured with prob = 1

X = learning

Y = Homework

LHS = $P(X \mid Y)$ is Prob of X given Y

$$P(Y \mid X) = \frac{P(X,Y)}{P(X)}$$

$$X \perp Y \iff P(X \mid Y) = P(X) \iff P(Y \mid X) = P(Y)$$

$$\Rightarrow \frac{P(X,Y)}{P(Y)} = P(X) \Rightarrow \boxed{P(X,Y) = P(X)\, P(Y)}$$

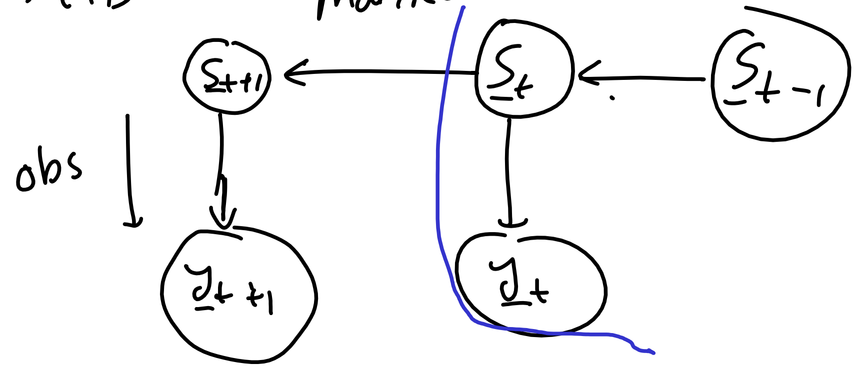
factorization

$$P(X, Y) = P(X|Y)\,P(Y) = P(Y|X)\,P(X)$$ | Always true

$$P(S_{t+1} \mid y_1, y_2 \ldots y_{t+1})$$

Localization/Mapping/State estimation

Markovian



Model

S_t blocks path from S_{t-1} to S_{t+1}

$$\left.\begin{array}{l} S_{t+1} \perp S_{t-1} \mid S_t \\ y_{t+1} \perp S_t \mid S_{t+1} \end{array}\right\}$$ Assumptions

$$P(x) = \int_y P(x, y)\,dy$$

$$P(S_{t+1} \mid y_1, y_2, \ldots, y_{t+1}) = \int_{S_t} P(S_{t+1}, S_t \mid y_1, y_2 \ldots y_{t+1})\,dS_t$$

$$\boxed{P(x, y \mid z) = P(x \mid y, z)\,P(y \mid z)}$$

$$\boxed{P(x, y) = P(x \mid y)\,P(y)}$$

$$= \int_{S_t} P(S_{t+1} \mid S_t, y_1 \ldots y_{t+1})\,\underbrace{P(S_t \mid y_1 \ldots y_{t+1})}_{}\,dS_t$$

$$= \int_{S_t} \underbrace{P(S_{t+1} \mid S_t, y_1, \ldots, y_{t+1})}_{\text{current time step}}\,\underbrace{P(S_t \mid y_1, \ldots, y_t)}_{\text{Prev time step}}\,dS_{t+1}$$

$$P(\underline{s}_{t+1} \mid \underline{s}_t, \underbrace{\underline{y}_1, \dots,}_{\underline{y}_1 \quad \underline{y}_t} \underline{y}_{t+1}) = P(\underline{s}_{t+1} \mid \underline{s}_t, y_{t+1})$$

Using

$$\underline{s}_{t+1} \perp \underline{y}_t \mid \underline{s}_t$$

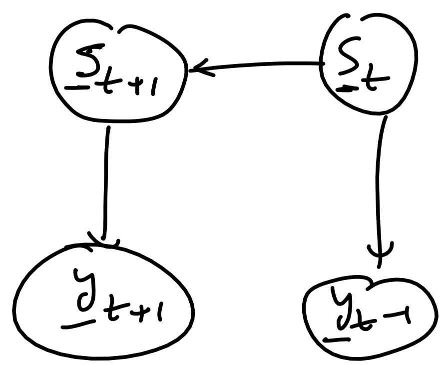
<u>Bayes theorem</u>

$$P(x, y) = \underbrace{P(x \mid y) P(y) = P(y \mid x) P(x)}$$

$$P(x \mid y) = P(y \mid x) \frac{P(x)}{P(y)} \qquad P(y \mid x) = P(x \mid y) \frac{P(y)}{P(x)}$$

$$P(\underline{s}_{t+1} \mid \underline{s}_t, \underline{y}_{t+1}) = \underbrace{P(\underline{s}_t, \underline{y}_{t+1} \mid \underline{s}_{t+1})} \frac{P(\underline{s}_{t+1})}{P(\underline{s}_t, \underline{y}_{t+1})}$$

$$= \underline{P(\underline{s}_t \mid \underline{s}_{t+1})} \; \underline{P(\underline{y}_{t+1} \mid \underline{s}_{t+1})} \frac{P(\underline{s}_{t+1})}{P(\underline{s}_t, \underline{y}_{t+1})}$$



What were the distributions given to us

① Motion model

$$P(\underline{s}_{t+1} \mid \underline{s}_t, \underline{a}_t) = \underline{P(\underline{s}_{t+1} \mid \underline{s}_t)} \qquad \underline{s}_t \perp \underline{y}_{t+1} \mid \underline{s}_{t+1}$$

② Observation model

$$\underline{P(\underline{y}_{t+1} \mid \underline{s}_{t+1})} = P(\underline{y}_t \mid \underline{s}_t)$$

$$P(\underline{s}_{t+1} \mid \underline{s}_t, \underline{y}_{t+1}) = P(\underline{s}_{t+1} \mid \underline{s}_t) P(\underline{s}_t) \quad P(\underline{y}_{t+1} \mid \underline{s}_{t+1}) \frac{1}{P(\underline{s}_t, \underline{y}_{t+1})}$$

$$\boxed{P(\underline{s}_{t+1} \mid \underline{s}_t, y_{t+1}) = \frac{P(\underline{s}_{t+1} \mid \underline{s}_t) P(\underline{y}_{t+1} \mid s_{t+1})}{P(y_{t+1} \mid s_t)}}$$

$$\frac{P(s_t, \underline{y}_{t+1})}{P(\underline{s}_t)} \rightarrow \underbrace{P(\underline{y}_{t+1} \mid s_t)}_{\text{Prior}}$$

common to assume a uniform prior

$$\boxed{P(\underline{s}_{t+1} \mid y_1 \ldots y_{t+1}) = \int_{s_t} \frac{\overbrace{P(\underline{s}_{t+1} \mid \underline{s}_t)}^{\text{motion model}} \overbrace{P(y_{t+1} \mid s_{t+1})}^{\text{obs model}}}{P(y_{t+1} \mid s_t)} P(\underline{s}_t \mid y_1, \ldots y_t)\, d s_t}$$

Bayes filter → particle filter

→ Kalman filter